

analyses any such they find.

[<http://setiathome.berkeley.edu>]

Non-profit organisations, universities, and even commercial organisations such as United Devices, Intel, IBM, Sun, and Microsoft are actively participating in scientific research. These are focused on designing distributed computing processes using open and scalable platforms on Internet-connected systems.

How It Works

So what we're talking about here is, you can help by "becoming part of a grid," donating the idle time of your computer. If you would like to help, there are a number of projects by the Berkeley Open Infrastructure of Network Computing (BOINC), Stanford University, World Community Grid, the National Foundation for Cancer Research, The Rothberg Institute for childhood diseases, and others.

The majority of the software clients for the purposes of these distributed computing projects are available for the Windows, Mac, and Linux/FreeBSD/Unix platforms. An application needs to be installed on the volunteer's computer, which will act as an agent and use that computer's CPU power when the computer is idle and connected to the Internet. After you download and install this software, they return the identification details of your system to the host computer server.

You'll then need to choose the projects you want to participate in, through this common software, for example, that of BOINC. As a result, when you're registered with a project, a screensaver runs when the computer is idle; the software works

on calculations in the background, and sends back the calculated results. Note that calculation is done only during idle time, so there's no interference with your regular work.

The Efforts Thus Far

Many projects are currently underway in various fields. Let's take a look at some of these.

BOINC-based Projects

BOINC is a distributed computing infrastructure that was developed after the success of SETI@home. BOINC serves as a software platform by which one can register for a number of distributed computing projects. Participation in multiple projects is possible through a single BOINC software.

BOINC was developed by a team based at the University of California at Berkeley, led by David Anderson, who works there as a research scientist. It is supported by the US National Science Foundation through awards for their research and development.

A number of projects come under the BOINC umbrella. These are in the fields of extraterrestrial intelligence, climate changes, chemistry, nanotechnology, cryptography, mathematics, and the life sciences. Here are some scientific research projects ongoing at BOINC. The current BOINC program version is 5.4.9, which is what you'll need to install. For more details, head to <http://boinc.bakerlab.org>.

Folding@home: This project, run by Stanford University, is a project where accurate simulations of protein folding and unfolding enable the scientific

community to better understand the development of many diseases, including Alzheimer's disease, BSE (mad cow disease), cancer, Huntington's Disease, cystic fibrosis, and other aggregation-related diseases.

Sony announced the Folding@home client for the PlayStation 3, which will contribute to medical research by utilising the idle time of the PS3's Cell processor. Recently, Stanford University announced the release of software that will utilise the processing power of ATI X1900 graphics processing units from ATI Technologies, and that will help them bring down the research period of Folding@home from three years to just a month. [<http://folding.stanford.edu>]

Climateprediction.net:

Under this project, the climate is being monitored so as to forecast climatic changes. The BBC Climate Change Experiment and Seasonal Attribution Project, which helps determine extreme weather events (cyclones, snowfalls, and the like) that could cause an impact on human activity, are also part of the project.

[<http://climateprediction.net>]

Rosetta@home: This one aims at solving the protein prediction problems run by the Baker Lab at the University of Washington. The ultimate goals of this project are the accurate prediction of human proteins that consist of amino acids, and the design of the protein structures that will help in the development of cures for diseases such as HIV/AIDS, cancer, Alzheimer's disease, malaria, and many others. The project has been successful thus far in accurate protein prediction.

[<http://boinc.bakerlab.org/rosetta>]

Jargon Buster

Gigaflops, Teraflops, Petaflops: FLOPS is an acronym for Floating Point Operations per Second. These are used as a measure of a computer's performance, especially while performing scientific calculations which make heavy use of floating-point calculations.

Aggregation-related diseases: Diseases caused due to clustering or amassing together of independent but similar particles, parts or bodies. For example, Alzheimer's disease is caused by development of abnormal tissues and protein deposits over thin surfaces consisting of nerve cell bodies.

Protein Prediction: A process of accurate prediction and design of protein structure and protein complexes.

Proteome Folding: The determination of the shape that certain proteins will take, and what kind of structure they will form, is called proteome folding. Proteome refers to the collection of all human proteins.

Protein Folding: A process by which proteins that consists of amino acids in the human body coil into a three-dimensional shape to perform their biological function.

Similar Ventures

World Community Grid: The WCG is run and funded by IBM to create the world's largest computing grid, and uses the idle time of volunteering Internet-connected computers. Currently, the WCG is undertaking research projects such as Help Defeat Cancer, Fight-AIDS@home, and Human Proteome Folding.

[www.worldcommunitygrid.org/projects_showcase/viewResearch.do]
Drug Design and Optimisation